Product requirements document: Personalized English Learning System

1 Vision & Strategic goal

A scalable English learning system that delivers **measurable up-skilling** for employees within a **personalized** learning journey while **eliminating manual bottlenecks** for account managers and giving HR clear **ROI visibility**.

Unique Value Proposition

Automated upskilling engine for specific business communication competencies. It delivers a measurable improvement in a single, high-impact skill through a hyper-personalized, AI-generated curriculum, providing transparent ROI with zero administrative overhead.

This UVP directly addresses the un-met needs of the sophisticated corporate buyer who has moved beyond asking "Are my employees learning English?" to asking "Are my employees getting better at negotiating deals in English, and can you prove it without creating more work for my team?"

2 Target Market & Business Model

Segment: Mid-to-large size firms (~50 – 2 000 employees)
Pricing: Per-course licence (individual / group) + paid add-ons (e.g. AI-Writing workshops)
Buying Persona: HR / L&D Manager who must prove ROI to leadership
Competitive Edge: Hyper-personalised curriculum with transparent analytics and low HR overhead

3 Core Values

Personalised
Transparent
Automated
Data-driven
Modern
No-bullshit

4 Success Metrics

Stakeholder	KPI	Target
Student	CEFR Delta in 6 months	+0.5 level in each skill
	NPS after 6 months	> 70
Teacher	Billable hours / week	+15% YoY
	NPS after 6 months	> 70
HR / Client	Churn rate in first 6 months	< 10%
	Dashboard adoption	> 80% log-in monthly
Account manager	Manual onboarding time	< 10 min. / learner

5 Jobs-to-Be-Done

Student - *"Give me lessons relevant to my job so I stay engaged and see progress."*
Teacher - *"Provide a pre-filled needs analysis and support so I can focus on teaching."*
HR - *"Show me live dashboards proving ROI with no extra admin."*
Account Manager - *"Automate folder creation, data collection, and progress reports so I spend less time per student."*

6 End-to-end learning journey

1. **Placement & Preference Form** (Forms → Sheet)
2. **Needs-Analysis Session** (AI-prefilled template; teacher confirms track & test band)
3. **Progress Speaking Test & AI Scoring** (16 versions = 4 skills × 4 bands)
4. **Teacher Validation — raise objection if score seems wrong (feeds retraining); otherwise accept.**
5. **AI-generated 24 h Course Plan** (8 blocks ≈ 6 months)
6. **Confirmation of Course Plan by Teacher and Student**
7. **Delivery** (60 min / week)
8. **Progress Speaking Retest & Dashboard update**

7 Pedagogical framework

7.1 Language Domains & Weighting

Domain	Scoring weights* MVP	Representation in course plan / lesson	Submetrics to include directly	Metrics to post-process with AI
Grammar	20%	40% explicit micro-targets	Grammatical accuracy & errors list, grammatical_range	noun_phrase_complexity, noun_phrase_variation, verb_construction_variation
Vocabulary	20%	40% explicit micro-targets	Word_sophistication, lexical_diversity	collocation_commonality, word_specificity
Pronunciation	20%	20%	2-3 lowest scoring phonemes with word / sentence examples	
Fluency	20%	0% - implicit	WCPM, articulation rate, all_pause_count & all_pause duration (for calculation of pauses per minutes)	
Coherence	20%	0% - implicit	Basic & causal connectives Adverb & verb diversity	

7.2 Skill-tracks

Learners pick exactly **one** skill-track to maximise focus and measurable depth. All test tasks and course-plan objectives must align to the learner's one selected track across the pipeline.

There are currently 4 thematic modules available:

1. **Meetings & Presentations** - Facilitate productive discussions and communicate with impact
2. **Business Negotiation** - Persuade diplomatically to reach agreements
3. **Building Relationships & Client Care** - Build professional connections and ensure satisfaction
4. **Leadership & Decision Making** - Lead teams effectively and navigate complex choices

Skill track reference

7.3 Course plan template

- 8 blocks × 3 h = 24 h
    - 21 h teaching
    - 3 h buffer / revision
- Each block includes "can-do" micro-goals aligned with CEFR.

- At least 3 blocks reflect student's hobbies and interests

7.4: Foundational Curriculum Mapping

The entire system's logic is predicated on the definitive alignment between the CEFR, the Global Scale of English (GSE), and the eight levels of the *Business Partner* coursebook series. This mapping is the foundational dataset and single source of truth for all test design and personalization algorithms.

8. Assessment framework

8.1 Band Routing

This table is the single source of truth for routing. Content for one test is sourced from multiple Business Partner books.

Test band	Source coursebooks	Target GSE range
A2	A1, A2, A2+, B1	22-42
B1	A2+, B1, B1+, B2	43-58
B2	B1+, B2, B2+, C1	59-75
C1	B2+, C1	68-88

Target GSE range = used for tasks creation and primary analysis

8.2 Progress Test Design

The system utilizes a library of 16 pre-defined static test versions (4 bands x 4 skill tracks). The creation of these tests is guided by a systematic framework to ensure relevance, appropriate difficulty, and diagnostic richness.

The logic for constructing each test version is as follows:

1. **Define Inputs:** For a given test version (e.g., B2 Band, Meeting Management), the core parameters are taken from the Band Routing table:
    - **Skill Track:** Meetings & Presentations
    - **Source Coursebooks:** *BP B1+, BP B2, BP B2+*
    - **Target GSE Range:** 59-75
    - **List of objectives**
2. **Select Task 1 - Structured Monologue**
3. **Select Task 2 - Spontaneous Dialogue / Roleplay**

4. **Finalize Test Package:** Each of the 16 test versions is finalized as a package containing:
    - Student's script for Task 1 and Task 2
    - Teacher's instructions (contains both student's script, teacher's script + additional instructions or hints)
    - A predefined list of the primary GSE descriptors each task is designed to elicit. This list is the direct input for the AI's Primary Analysis

Task selection criteria are Format, GSE Level, and Skill Track alignment. However, where multiple suitable task pairs exist that meet these criteria, the framework should prioritize pairs that are thematically connected to create a more cohesive and natural test flow for the learner. Thematic connection is considered a preferential, not a mandatory, attribute..

9.0 AI Evaluation and Scoring

9.1 AI Logic for Test Evaluation and Course Plan Generation

The AI engine uses a dual-analysis model to transform the speaking test results into a personalized course plan.

Data Ingestion: To perform a comprehensive analysis, the AI engine ingests a structured set of data for each learner. This ensures that the subsequent analysis is grounded in the full context of the learner's performance, goals, and the system's pedagogical framework.

The AI receives the following inputs:

- **Core Test Data:**
    - **Proficiency Anchor:** The final, holistic GSE score (e.g., 62) generated by Speechace from the entire spoken performance.
    - **Diarized Transcript:** The complete transcription of the speaking test, including both the teacher's prompts/questions and the student's responses for Task 1 and Task 2.
    - **Test Version ID:** The identifier for the specific test administered (e.g., "B2_Meeting_Management_v1"), which links to the targeted GSE descriptors for the Primary Analysis.
- **Learner Profile Data:**
    - **Selected Skill Track:** The single skill track chosen by the learner (e.g., "Meeting Management").
    - **Learner Interests:** A list of hobbies and interests provided by the learner in their preference form.
- **System Knowledge & Templates:**
    - **Curriculum Mapping Database:** A structured database that links each of the four skill tracks to a curated list of specific Business Partner lessons (e.g., "A2_U5.1") relevant for each of the four test bands. This mapping is the primary source for:
        - Test Construction: Identifying relevant scenarios and content for the 20 static test versions.

        - Course Plan Generation: Selecting appropriate lessons and activities from the Business Partner series to populate the 8-block course plan based on the AI's diagnostic findings.
    - **GSE Descriptor Database:** Full, structured access to the database of all GSE learning objectives from the *Business Partner* series. Each descriptor includes its GSE score, "Can-Do" statement, and mapping to relevant skill tracks.
    - **Course Plan Template:** The standardized, 8-block structure that the AI is required to populate to generate the final 24-hour course plan.

AI steps:

1. Primary Analysis: Performance on Targeted Skills

This analysis evaluates performance against the skills the test was explicitly designed to measure.

- **Logic:**
    1. Each test version has a predefined list of *targeted* GSE descriptors. These descriptors fall within the test's **Target GSE Range** (e.g., 59-75 for the B2 test). The number of objectives varies but is always between 10 and 14.
    2. The AI analyzes the diarized transcript for evidence of the learner attempting to perform the functions described by these targeted descriptors.
    3. Each targeted descriptor is classified as either **"Demonstrated"** (clear, successful evidence found), **"Partially Demonstrated"** or **"Not Demonstrated"** (attempt failed or avoided).
- **Output:**
    1. A high-priority list of **"Not Demonstrated"** skills, representing the learner's most critical gaps.

2. Secondary Analysis: Spontaneous Performance in Dynamic Proximal Range (DPR)

This analysis searches for spontaneously produced skills to identify emergent abilities and ideal "stretch goals". To create a candidate pool that is both robust and logically scaled to the task, the system calculates the number of objectives for review based on the number of lessons remaining in the course plan.

Logic:

1. **Calculate Need (R):** The system first determines the number of remaining lesson slots (R) to be filled in the 14-hour lesson plan. If R is 0, this analysis is skipped.
2. **Calculate Number of Objectives to Select (N):** The number of high-relevance objectives to select (N) is calculated using a simple linear formula:

$$N = 10 + (R * 2)$$

This ensures the candidate pool scales directly with the number of lessons needed, with a minimum of 12 objectives for **R=1** and a maximum of 38 for **R=14**.

3. **Filter, Sort, and Slice:** The system performs the following sequence:
    - Queries the GSE learning objectives database to find all objectives that align with the learner's single selected skill track and fall within their Dynamic Proximal Range (DPR).
    - Sorts the resulting list of objectives by skill track relevance score in descending order.
    - Selects the **top N** objectives from this sorted list.
4. **AI Transcript Analysis:** This final list of N high-relevance objectives is passed to the AI for analysis against the diarized transcript.
5. **Classification:** Each candidate objective is classified as either **"Observed"** (positive evidence found) or **"Not Observed"** (no evidence found).

Output: A secondary-priority list of "Not Observed" skills, representing ideal stretch goals. The lessons associated with these skills are then used to populate the remaining slots in the course plan.

3. Synthesis and Course Plan Generation

The AI populates the 24-hour course plan using following logic

- 3 hours are allocated as buffer / time for revision
- 7 hours are allocated as a hobby / downtime / rest hours to maximize engagement
- Remaining 14 hours are lessons oriented towards mastering objectives
    - One lesson corresponds to one book reference
    - Each learning objective has it's associated course reference. Most of them has single reference, some of them multiple. If objective has more than 2 references, we would only count 2 references toward filling the hours
    - Logic is following:
        - 1. Learning objectives labeled as **"Not Demonstrated"** and **"Partially demonstrated"**
        - **2**. If there are any lessons left, add learning objectives from secondary analysis which were **"Not Observed"**

9.2 Holistic Scoring and the Diagnostic Hierarchy

The system employs a hierarchical model to ensure scoring is both objective and diagnostically rich.

1. Principle of Holistic Scoring: The entire spoken performance (Task 1 and Task 2) is submitted to the automated scoring engine Speechace for analysis:

- The audio is first transcribed and diarized via AssemblyAI.
- Speechace returns numeric domain scores and overall score per each utterance
- Speechace returns utterance-level metrics and numeric domain scores (0–100) + overall score.
- The final overall GSE score is calculated as a duration weighted average of the overall scores per utterance
- Final GSE score is converted to CEFR

Notes: Sub-category metrics are stored for diagnostics only and do not influence scoring weights. Whole Speechace output is stored in case we want to use it in future (e.g. we will display CEFR scores in the dashboard)

2. The Proficiency Anchor (Speechace): Final GSE score is the **Proficiency Anchor** and is the definitive, quantitative measure of the learner's overall proficiency level.

3. The Diagnostic Detail (Our AI): Our AI's function is to provide the qualitative diagnosis that explains the Proficiency Anchor score. It takes the final GSE score and the full transcript as inputs to perform its Primary and Secondary Analyses.

4. Resolving Data Conflicts: A perceived conflict between the Speechace score and our AI's analysis is not an error, but the core of the diagnosis. For example, if Speechace assigns a B2 score but our AI identifies a "Not Demonstrated" B1-level grammar point, the system's interpretation is: "The learner's overall profile is B2, but a specific B1-level gap is preventing them from being a more solid B2 speaker." This B1 skill is then prioritized in the course plan.

10. Technical architecture

10.1 Stack

- Google Workspace
- Apps Script
- n8n / Make
- ChatGPT API
- Looker Studio
- Firebase authentication (optional)

10.2 Automation touch-points

Trigger	Script	Task	Output
Form Submit	App Script	Create client / student folder	Drive hierarchy

Student folder created	Make / n8n + AI	Script - pre-fill Needs Analysis	Needs analysis pre-filled doc
Needs Analysis filled	App Script	Construct speaking test	Speaking test doc
Audio upload	n8n + AI	Score	Speaking test results doc
Score approval	App Script + AI	Generate course plan	Google Sheet
Retest submit	App Script	Progress delta + dashboard push	Bigquery / Looker refresh

11. MVP Scope & Phases

MVP - Core Automation

- Form → folder & needs analysis pre-fill
- Speaking test AI scoring
- Course plan generator

Post-MVP

- AI evaluations to improve precision
- Looker dashboard
- 6-month comparison

12. Components overview

Needs analysis

Needs Analysis builds on Preference & Placement test data through structured conversation covering learning preferences, work challenges, and systematic track selection to create student commitment for a 6-month journey.

Section	Content	Teacher's action	When to perform
Pre-lesson preview	Test results - CEFR, total score and subscores, test link Background - job, pre-selected skill track, hobbies, favorite resources	Review	Before lesson
Warm-up	AI generated hobby questions for	Review	Before and

	warm-up and level assessment	Refine Ask	during lesson
Learning context	Learning experience, activities, preferred lesson types, homework	Ask Fill	During lesson
Explaining the learning journey	Learning journey process outline	Explain	During lesson
Work Context & Skill track selection	Work situation & needs, Final selection of skill track(s).	Fill Help to choose	During lesson
Band routing	Select appropriate level (A1-A2, B1, B2, C1-C2)	Select	During lesson
Teacher notes	Key observations, other notes	Fill	During / after lesson